Notes: Chang, Wang, and Xiong (RFS 2026)

Price and Volume Divergence in China's Real Estate Markets: The Role of Local Governments

1 What the paper does

The paper documents a striking **price–volume divergence** in China's real estate markets during 2020–2022: *land and new-housing prices keep rising* while *transaction volumes fall*. The central thesis is that this divergence reflects **active local-government market management**.
The paper emphasizes three related channels:

- **Supply management:** local governments restrict effective land supply (and, in housing markets, reduce the issuance of sales permits), helping prevent prices from falling even when demand weakens.
- **LGFV[1] interventions:** LGFVs step in as countercyclical buyers in land auctions and (in some periods) bid more aggressively, supporting prices and transaction benchmarks.
- **Debt-financing incentives:** supporting land prices helps preserve land-sale revenue and sustains land-backed (implicit) local-government debt financing; this shows up in bond-market evidence linking higher land prices to lower corporate financing costs.

Empirically, the paper combines transaction-level land data, city-year panels, and firm/bond outcomes, and uses **cross-city heterogeneity in pre-COVID fiscal exposure** (land dependence and LGFV debt, both measured in 2019) to identify stronger local-government price support where incentives are larger.

2 Data

- Residential land transactions from China Land Market website: 133,971 parcels in 173 cities (2015–2022).
- New-housing prices/transactions from CREIS; housing sales permits (area and units).
- City fiscal variables: land dependence and LGFV debt (both measured using 2019 levels to define pre-COVID fiscal conditions).
- Bond-market sample: 31,316 bonds by 2,797 firms; matched to city-level land prices.

[1]In this paper, *LGFV* stands for *local government financing vehicle* (often called an "urban construction and investment company"). Authors use the term to mean a firm that is owned or backed by a local government and is used as a financing conduit because Chinese budget law restricts direct local-government borrowing. LGFVs borrow from banks and capital markets with implicit government support, and historically they financed land development and infrastructure by pledging land or future land-sale proceeds as collateral; over time, many LGFVs expanded into real-estate development/commercial operations and also participate directly in land auctions. Although LGFV liabilities are not legally explicit local-government debt, they are widely treated as off-balance-sheet public debt, and lenders often require collateral based on market values—which ties LGFVs' refinancing capacity (and thus local governments' fiscal room) to observed land and housing prices.

3 Research questions and identification

The authors are asking (and answering) a tightly connected set of questions:

1. **Fact / measurement:** Did China experience a sustained *price–volume divergence* in (i) the *residential land market* and (ii) the *new-housing market* during the COVID-19 period (2020–2022)? In other words: did prices remain high (or rise) while transaction volumes fell? (Figure 1; Tables 2–4 and Table 10.)
2. **Economic mechanism:** If such a divergence exists, is it consistent with *local governments' strategic price management* rather than standard private-market frictions? The key idea is that sellers are institutional and policy-driven (local governments sell land; developers sell new homes), so typical homeowner loss-aversion stories are less plausible.
3. **Cross-sectional incentives:** Are the divergence and price increases *stronger in cities with stronger ex ante fiscal/debt incentives* to defend land/housing prices? The paper uses two 2019 (pre-COVID) city characteristics:
   - **Land dependence_2019:** land-sale revenue / total fiscal revenue in 2019.
   - **LGFV debt_2019:** aggregate LGFV liabilities / total fiscal revenue in 2019.

   The question is whether these pre-pandemic exposures predict larger post-2020 land-price increases and stronger price–volume divergence. (Table 5; Table 6.)
4. **Implementation channels:** *How* do local governments operationally support prices? The paper highlights:
   - LGFVs buying more land (quantity channel) (Table 7),
   - LGFVs bidding more aggressively / paying relatively higher prices in downturn (price channel) (Table 8),
   - restricting new-home supply via fewer sales permits in high-debt cities, and administrative price floors for new homes (Table 11 + narrative evidence).
5. **Consequences:** What are the broader implications of price management? The paper looks at (i) the land-price–financing-cost relationship for local firms (Table 12; interpreted as changing correlation, not clean causal), and (ii) developers' housing sales declines being larger in high-LGFV-debt cities (Table 13).

3.1 Identification strategy: what variation identifies the key claims?

The core identification is a **dynamic difference-in-differences / event-study design** that treats the COVID-19 period (2020–2022) as a large, plausibly exogenous negative shock, and compares outcomes across cities with different *pre-pandemic* exposure to land-based fiscal/debt incentives.

Step 1: Establish the price–volume divergence as a reduced-form fact. They first show that (i) land prices rise relative to 2019 even in 2021–2022 (Table 2), while (ii) land transaction volume collapses in 2022 (Table 3), and (iii) failure rates of land auctions rise (Table 4), suggesting the volume decline is not just a benign "supply shortage" story. They replicate a similar divergence in new-housing prices vs. transaction volume (Table 10). This part is primarily descriptive, but done with fixed effects and controls.

Step 2: Dynamic DID for land prices using *predetermined* 2019 exposures. The main causal-style evidence uses specifications of the form (their Eq. (3)):

$$\ln(\text{LandPrice})_{i,j,t} = \sum_{s\neq 2019} \alpha_s \big(\text{Exposure}_{j,2019} \times 1\{t=s\}\big) + \text{controls}_{j,t} + \text{controls}_{i,t} + \text{City FE}_j + \text{Time FE}_t + \varepsilon_{i,j,t}.$$

Here, the **"treatment intensity"** is either $\text{LandDependence}_{j,2019}$ or $\text{LGFVDebt}_{j,2019}$, measured *before* the shock, and the omitted baseline year is 2019. The coefficients α_s trace out whether higher-exposure cities have differential price movements in each year s relative to 2019.

- **Key identifying assumption (parallel trends):** Absent the pandemic shock, cities with different 2019 land dependence / LGFV debt would have had *similar* trends in land prices (after conditioning on city fixed effects and time-varying controls).
- **How they assess the assumption:** They use the *pre-2020* interaction coefficients as pre-trend tests. In Table 5, the interaction terms for years 2015–2018 are generally statistically insignificant, which they interpret as supportive evidence for parallel pre-trends.
- **What they find post-shock:** Interaction terms for 2021 and especially 2022 are positive and significant, implying that *higher 2019 exposure* predicts *larger* land-price increases in 2021–2022 (Table 5). When both exposures enter jointly (Table 5, col. 5), the LGFV-debt interactions remain strong, suggesting the debt/collateral channel is particularly important.

Step 3: Identification of the "divergence" cross-sectionally. To link the story directly to price–volume divergence, they run city-year regressions of land price growth on land area growth and interact land area growth with a COVID dummy (their Table 6):

$$\Delta P_{j,t} = a_1 \Delta Q_{j,t} + a_2(\Delta Q_{j,t} \times \text{Covid19}_t) + \cdots$$

The identifying variation is whether the *price–volume slope* changes sign during 2020–2022, and whether that change is concentrated in high-LGFV-debt cities (Table 6 shows the strongest negative divergence in the high-debt group).

Step 4: Identification of mechanisms (LGFV buying and bidding). They then show two complementary pieces of evidence tying local-government incentives to market behavior:

- **LGFVs buy more land during COVID (city-level event study):** They regress LGFV purchase shares on year dummies with city FE and controls (Table 7). The identifying variation is the within-city change in LGFV share over time, especially the 2022 jump, and the stronger increases in high land-dependence / high-debt city groups.
- **LGFVs pay relatively higher prices during COVID (transaction-level dynamic DID):** They estimate a transaction-level event-study with an LGFV-buyer indicator interacted with year dummies (Table 8). Identification comes from comparing the *relative* prices paid by LGFVs vs. non-LGFVs over time, within cities, controlling for parcel characteristics (FAR, land grade, urban location, etc.) and fixed effects. The key pattern is: LGFVs paid less than others in 2019 (negative level effect), but the relative price flips upward by 2022 (positive interaction), consistent with "buyer-of-last-resort" / price-support bidding.

Step 5: Identification in the new-housing market (permits channel). For housing prices and permit issuance, they again use a DID/event-study style approach where $\text{LGFVDebt}_{j,2019}$ is interacted with year dummies (Table 11). Identification relies on the same parallel-trends logic: conditional on FE and controls, high- vs. low-debt cities would have had similar trends absent the shock. They find high-debt cities have (i) larger house-price increases in 2021–2022 and (ii) fewer permits in 2022, consistent with supply restriction as a policy lever.

Step 6: "Impacts" analyses: what is (and is not) identified.

- **Firm financing (Table 12):** Here I do *not* read the paper as claiming a clean causal effect of land prices on financing costs, because local conditions jointly move land prices and spreads. The authors explicitly frame it as examining how the *correlation* between land prices and coupon spreads changes during COVID (interaction with the COVID dummy), which is suggestive about collateral/signal usage under stress, but not a sharp causal design.
- **Developers' sales (Table 13):** They use an event-study DID at the developer×city level with $\text{LGFVDebt}_{j,2019}$×year interactions (their Eq. (5)). Identification is again "differential post-shock effects by predetermined city exposure," with pre-trend checks (2015–2018 interactions insignificant) and extensive FE (developer, city, year). The finding is that developers' sales fall *more* in high-debt cities in 2022, consistent with the idea that price-support policies choke off transactions and worsen developer liquidity.

3.2 Main identification

The paper's main identification is: *treat COVID-19 as an exogenous negative shock and use a dynamic DID/event-study to show that cities with higher pre-2020 reliance on land-based finance (especially LGFV debt) exhibit larger post-2020 price increases, stronger price–volume divergence, and more LGFV intervention / supply restriction, consistent with strategic local-government price management.*

4 Empirical models in the paper (all main specifications)

Throughout, they treat 2019 as the omitted/baseline year (unless otherwise stated), and cluster standard errors as in the paper.

4.1 Transaction-level land price dynamics (Eq. 1)

For land parcel i in city c and year t:

$$\ln(\text{LandPrice})_{i,c,t} = \alpha_c + \sum_{k\in\{2015,2016,2017,2018,2020,2021,2022\}} \beta_k \cdot \mathbb{1}[t=k] + \Gamma^\top X_{i,c,t} + \varepsilon_{i,c,t}. \qquad (1)$$

Interpretation. The year-dummy coefficients β_k trace the land-price path relative to 2019 (conditional on fixed effects and controls). $X_{i,c,t}$ includes land characteristics (e.g., ln(land area), FAR, Urban, New land, Land grade, and transaction format).

4.2 City-year panel dynamics for transaction volumes (Eq. 2)

For city c in year t:

$$Y_{c,t} = \alpha_c + \sum_{k\in\{2015,2016,2017,2018,2020,2021,2022\}} \beta_k \cdot \mathbb{1}[t=k] + \Theta^\top Z_{c,t} + u_{c,t}, \qquad (2)$$

where $Y_{c,t}$ is typically ln(city land area) or ln(city land value) (and later analogs for housing), and $Z_{c,t}$ are city economic/fiscal controls.

4.3 Dynamic DID with pre-determined fiscal exposure (Eq. 3)

Let E_c be a pre-COVID (2019) exposure measure: either `Land dependence_2019` or `LGFV debt_2019`. Then:

$$\ln(\text{LandPrice})_{i,c,t} = \alpha_c + \alpha_t + \sum_{k\in\{2015,2016,2017,2018,2020,2021,2022\}} \delta_k \cdot (E_c \times \mathbb{1}[t=k]) + \Theta^\top Z_{c,t} + \Gamma^\top X_{i,c,t} + \varepsilon_{i,c,t}. \tag{3}$$

Interpretation. The interaction coefficients δ_k tell me whether higher-exposure cities experienced *differential* land-price dynamics in year k (relative to 2019). Key identifying idea: using 2019 exposure makes it plausibly predetermined with respect to the COVID-era shock.

Dynamic DID for LGFV bidding behavior (variant used for Table 8)

A closely related transaction-level design replaces the city exposure E_c with an indicator for whether the parcel is bought by an LGFV:

$$\begin{aligned} \ln(\text{LandPrice})_{i,c,t} = {} & \alpha_c + \alpha_t + \beta\, \text{LGFVdummy}_{i,c,t} \\ & + \sum_{k\in\{2015,2016,2017,2018,2020,2021,2022\}} \theta_k \cdot \Big(\text{LGFVdummy}_{i,c,t} \times \mathbb{1}[t=k]\Big) \\ & + \Gamma^\top X_{i,c,t} + \Theta^\top Z_{c,t} + \varepsilon_{i,c,t}. \end{aligned} \tag{4}$$

Interpretation. β captures how LGFVs bid relative to other buyers in the omitted year (2019). The interaction coefficients θ_k capture how the LGFV vs. non-LGFV bid gap changes in each year k.

4.4 Price–volume divergence as a (negative) correlation during COVID (Eq. 4)

At the city-year level:

$$\text{LandPriceGrowth}_{c,t} = \alpha_c + \alpha_t + a_1\, \text{LandAreaGrowth}_{c,t} + a_2\, (\text{LandAreaGrowth}_{c,t} \times \texttt{Covid19}_t) + \Theta^\top Z_{c,t} + \eta_{c,t}, \tag{5}$$

where $\texttt{Covid19}_t = 1$ for 2020–2022 and 0 otherwise. **Interpretation.** Before COVID, a_1 captures the price–volume comovement. During COVID, the slope becomes $a_1 + a_2$. A negative a_2 (and negative $a_1 + a_2$) is evidence of divergence.

4.5 Land prices and firm financing costs (Eq. 5)

For bond i issued by firm f in city c at time t:

$$\begin{aligned} \text{CouponSpread}_{i,f,c,t} = {} & \alpha_c + b_1 \ln(\text{LandPriceAvgQ})_{c,t-1} + b_2 \Big(\ln(\text{LandPriceAvgQ})_{c,t-1} \times \texttt{Covid19}_t\Big) \\ & + \Phi^\top W_{i,f,t} + \mu_f + \lambda_t + \varepsilon_{i,f,c,t}. \end{aligned} \tag{6}$$

Interpretation. If local governments support land prices to protect their debt-financing capacity, I expect higher land prices to be associated with *lower* financing costs (smaller coupon spreads), especially in the COVID period (captured by b_2).

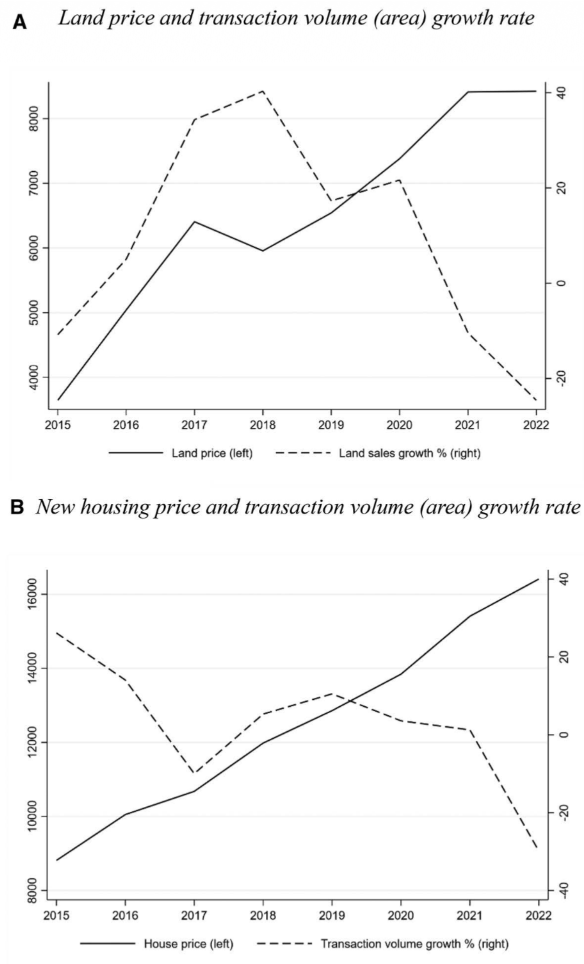

Figure 1
Price and volume divergence in real estate markets

4.6 Model–table

- Table 2: Eq. (1) (transaction-level land prices).
- Table 3: Eq. (2) (city-year land quantities/values).
- Table 4: Eq. (2) with failure ratios as outcomes.
- Table 5: Eq. (3) (dynamic DID with 2019 land dependence / LGFV debt).
- Table 6: Eq. (4) (price-growth on volume-growth slope shift during COVID).
- Table 7: Eq. (2) with LGFV purchase shares (area/value) as outcomes.
- Table 8: Eq. (3') (LGFV bidder DID).
- Table 9: Eq. (2) with LGFV leverage as outcome.
- Table 10: Eq. (2) with housing price and housing transaction volumes as outcomes.
- Table 11: Eq. (3) with housing price / permit issuance as outcomes (exposure = LGFV debt_2019).
- Table 12: Eq. (5) (bond spreads on lagged city land prices, with COVID interaction).
- Table 13: Eq. (3) style DID at the developer×city×year level (outcome = developer housing sales).

Table 1
Summary statistics

Variable	N	Mean	SD	p5	p25	p50	p75	p95
				A. Transactions involving new housing units at the city level				
ln(house price)	1,052	9.088	0.479	8.525	8.748	8.985	9.308	10.020
ln(house transaction units)	1,053	10.129	1.026	8.472	9.464	10.079	10.828	11.813
ln(house transaction area)	1,053	14.863	1.019	13.210	14.215	14.822	15.587	16.503
ln(house transaction value)	1,072	14.743	1.255	12.806	13.931	14.593	15.659	16.916
ln(permit area)	706	14.861	1.216	12.824	14.150	14.852	15.763	16.640
ln(permit units)	705	10.077	1.344	7.968	9.393	10.118	11.024	11.889
				B. Residential land transactions at the transaction level				
ln(land price)	133,965	7.897	1.064	6.217	7.202	7.873	8.591	9.661
ln(land area)	133,971	1.228	0.743	0.013	0.607	1.325	1.806	2.350
LGFV dummy	133,971	0.168	0.374	0	0	0	0	1
FAR	129,690	2.496	1.159	1.111	1.800	2.200	2.917	4.900
Urban	133,971	0.401	0.490	0	0	0	1	1
New land	133,971	0.505	0.500	0	0	1	1	1
Land grade	125,234	5.744	4.453	1	2	4	10	14
Tender	133,971	0.004	0.061	0	0	0	0	0
Auction	133,971	0.287	0.452	0	0	0	1	1
				C. Residential land transactions at the city level				
ln(city land area)	1,384	5.406	1.016	3.694	4.860	5.549	6.114	6.751
ln(city land value)	1,384	13.625	1.728	11.314	12.861	13.718	14.540	15.973
ln(city land supply area)	1,375	5.125	1.121	2.971	4.551	5.306	5.924	6.556
ln(city land supply piece)	1,375	4.049	1.041	2.079	3.466	4.159	4.796	5.521
Land price growth rate	1,374	0.163	0.426	−0.348	−0.101	0.076	0.322	0.921
Land area growth rate	1,375	0.099	0.582	−0.578	−0.285	−0.003	0.309	1.205
ln(land price_avgq)	4,049	8.375	0.849	7.059	7.832	8.292	8.894	9.953
Failure ratio (area)	1,374	0.107	0.131	0	0	0.056	0.171	0.388
Failure ratio (piece)	1,374	0.108	0.126	0	0	0.060	0.177	0.377
LGFV land ratio (area)	1,383	0.168	0.169	0	0.026	0.118	0.263	0.523
LGFV land ratio (value)	1,383	0.159	0.174	0	0.018	0.094	0.243	0.551
LGFV debt_2019	173	1.533	1.396	0.172	0.490	1.037	2.316	4.614
Land dependence_2019	152	0.381	0.130	0.141	0.294	0.392	0.473	0.586
				D. Economic indicators at the city level				
ln(GDP per capita)	1,355	11.073	0.475	10.306	10.725	11.069	11.406	11.881
GDP growth	1,383	0.064	0.028	0.013	0.044	0.071	0.083	0.097
Fiscal deficit	1,381	0.094	0.068	0.014	0.045	0.083	0.124	0.209
Tax ratio	1,359	0.722	0.086	0.580	0.663	0.722	0.785	0.857
Secondary sector	1,377	0.422	0.084	0.268	0.374	0.423	0.481	0.554
Tertiary sector	1,377	0.485	0.085	0.363	0.429	0.475	0.526	0.655
LGFV leverage	1,328	52.798	9.737	35.275	47.222	53.643	59.077	66.881
				E. Bond market indicators				
Coupon spread	31,316	1.318	1.114	0.010	0.553	1.028	1.835	3.695
ln(issue amount)	31,316	6.702	0.720	5.670	6.215	6.802	7.090	8.006
Maturity	31,316	1.591	1.670	0.164	0.493	0.740	3.000	5.000
TripleA	31,316	0.583	0.493	0	0	1	1	1
DoupleAplus	31,316	0.282	0.450	0	0	0	1	1
SOE	31,316	0.914	0.280	0	1	1	1	1
ln(asset)	31,300	11.458	1.273	9.543	10.533	11.337	12.301	13.674
ln(sales)	31,195	9.702	1.871	6.777	8.216	9.655	11.188	12.714
Leverage	31,300	0.642	0.122	0.416	0.575	0.654	0.723	0.832
ROA	31,299	0.017	0.020	0.000	0.005	0.011	0.025	0.056
				F. Real estate developer indicators				
ln(house sales value)	16,898	10.935	2.009	6.669	10.006	11.298	12.321	13.588
ln(house sales area)	16,898	10.642	1.895	6.452	9.851	11.042	11.939	13.055
ln(asset)	1,286	10.691	1.598	8.065	9.644	10.705	11.877	13.331
ln(sales)	1,289	9.022	1.789	6.091	7.933	9.033	10.175	12.206
Leverage	1,286	0.710	0.171	0.333	0.639	0.750	0.824	0.907
ROA	1,286	0.019	0.033	−0.039	0.011	0.020	0.033	0.062

Figure

Figure 1A (land): Land prices rise strongly from 2019 to 2022, while land transaction-volume growth turns sharply negative in 2021–2022. This is the core land-market divergence.
Figure 1B (housing): New-housing prices continue rising, but new-housing transaction-volume growth collapses (especially in 2022). This mirrors the land-market pattern and motivates linking the divergence to local-government behavior.

Tables

Interpretation (Table 1).

- Land transactions: mean ln(land price) = 7.897 with substantial dispersion (SD = 1.064), and about 16.8% of parcels are bought by LGFVs (LGFV dummy mean = 0.168).
- City-level land dependence in 2019 averages 0.381 (large cross-city heterogeneity), and LGFV debt in 2019 averages 1.533 (liabilities around 1.5× fiscal revenue on average).
- Housing: prices (ln) look much smoother than volumes; permits are observed for a smaller sample (N ≈ 700 city-years).

Table 2
Land price dynamics

Variables	(1) *ln(land price)*	(2) *ln(land price)*	(3) *ln(land price)*
Year2015	–0.501***	–0.559***	–0.535***
	(–20.41)	(–10.74)	(–13.94)
Year2016	–0.439***	–0.470***	–0.476***
	(–18.94)	(–11.80)	(–14.00)
Year2017	–0.201***	–0.224***	–0.229***
	(–7.47)	(–6.49)	(–7.98)
Year2018	–0.078***	–0.089***	–0.082***
	(–4.03)	(–4.51)	(–4.73)
Year2020	0.070***	0.089***	0.054**
	(4.22)	(3.15)	(2.29)
Year2021	0.106***	0.152***	0.127***
	(5.22)	(5.88)	(5.64)
Year2022	0.117***	0.173***	0.110***
	(5.58)	(4.59)	(3.46)
ln(GDP per capita)		–0.140	–0.012
		(–1.52)	(–0.16)
GDP growth		0.233	–0.326
		(0.51)	(–0.86)
Fiscal deficit		0.867	–0.071
		(1.47)	(–0.14)
Tax ratio		0.317	0.131
		(1.56)	(0.84)
Secondary sector		0.293	0.471
		(0.30)	(0.51)
Third sector		–0.168	0.288
		(–0.17)	(0.32)
ln(land area)			0.083***
			(6.41)
FAR			0.241***
			(24.47)
Urban			0.519***
			(19.62)
New land			–0.122***
			(–9.59)
Land grade			–0.021***
			(–8.33)
Tender			0.036
			(0.36)
Auction			0.260***
			(9.39)
Constant	8.000***	9.193***	6.966***
	(392.29)	(6.97)	(6.33)
Observations	133,965	129,926	117,442
Adj. R-squared	0.384	0.385	0.536
City FE	YES	YES	YES

$$\ln(\text{LandPrice})_{i,j,t} = \alpha_0 + \sum_{\substack{s=2015\\ s\neq 2019}}^{2022} \alpha_s\,\text{Year}_{s,t} + \beta\,\text{CityControls}_{j,t} + \gamma\,\text{ContractControls}_{i,t} + \text{Fixed effects} + \varepsilon_{i,j,t}.$$

Interpretation (Table 2).

- Relative to 2019, land prices were much lower in 2015–2018 (large negative coefficients).
- **During COVID (2020–2022), land prices rise**: even with full controls (col. 3), Year2021 and Year2022 are strongly positive (0.127 and 0.110).
- Controls behave as expected: larger parcels and higher FAR are more expensive; urban locations command a premium; auction transactions are pricier than baseline; "new land" is discounted.

$$\text{Dep}_{j,t} = a_0 + \sum_{\substack{s=2015\\ s\neq 2019}}^{2022} a_s\,\text{Year}_{s,t} + \gamma^\top \textbf{CityControls}_{j,t} + \text{FE}_j + \text{FE}_t + \varepsilon_{j,t}. \tag{7}$$

Interpretation (Table 3).

- Transaction volume (area) falls sharply in 2022 (about -0.43 log points relative to 2019), while land prices are *higher* in 2022. This is the land-market divergence.
- 2020 shows an increase in volumes (positive coefficients), consistent with a rebound/boost in activity even as the pandemic began.
- In 2021, quantities do not collapse yet (coefficients near zero), but the big drop arrives in 2022.

Table 3
Dynamics of the land transaction volume

VARIABLES	(1)	(2)	(3)	(4)
	ln(city land area)		*ln(city land value)*	
Year2015	−0.440***	−0.379***	−1.064***	−1.033***
	(−8.39)	(−4.11)	(−15.74)	(−8.06)
Year2016	−0.530***	−0.459***	−0.981***	−0.909***
	(−12.63)	(−6.52)	(−16.84)	(−9.32)
Year2017	−0.304***	−0.285***	−0.449***	−0.432***
	(−7.04)	(−5.19)	(−7.26)	(−5.96)
Year2018	−0.092***	−0.114***	−0.144***	−0.178***
	(−3.22)	(−3.15)	(−3.71)	(−3.09)
Year2020	0.136***	0.168***	0.207***	0.259***
	(4.75)	(3.86)	(5.78)	(4.62)
Year2021	−0.048	−0.037	0.045	0.096*
	(−1.24)	(−0.80)	(0.89)	(1.67)
Year2022	−0.420***	−0.431***	−0.456***	−0.407***
	(−8.67)	(−6.38)	(−7.46)	(−4.88)
ln(GDP per capita)		0.101		−0.131
		(0.41)		(−0.39)
GDP growth		−0.008		0.040
		(−0.01)		(0.04)
Fiscal deficit rate		−0.136		−0.257
		(−0.08)		(−0.09)
Tax ratio		0.720*		1.326**
		(1.82)		(2.56)
Secondary sector		3.428		4.105
		(1.61)		(1.58)
Tertiary sector		3.310		4.291
		(1.54)		(1.60)
Constant	5.619***	0.937	13.980***	10.689**
	(231.58)	(0.32)	(449.66)	(2.38)
Observations	1,384	1,333	1,384	1,333
Adj. R-squared	0.829	0.833	0.892	0.887
City FE	YES	YES	YES	YES

Table 4
Dynamics of the failure ratio

VARIABLES	(1)	(2)	(3)	(4)
	Failure ratio (piece)		*Failure ratio (area)*	
Year2015	−0.090***	−0.052**	−0.086***	−0.064***
	(−10.98)	(−2.51)	(−10.51)	(−3.09)
Year2016	−0.089***	−0.065***	−0.083***	−0.070***
	(−11.77)	(−3.83)	(−10.50)	(−3.97)
Year2017	−0.068***	−0.048***	−0.066***	−0.054***
	(−8.44)	(−3.78)	(−8.93)	(−4.30)
Year2018	−0.025***	−0.016	−0.008	−0.001
	(−3.26)	(−1.61)	(−0.92)	(−0.06)
Year2020	0.060***	0.048***	0.053***	0.041***
	(6.22)	(3.64)	(5.51)	(3.09)
Year2021	0.112***	0.090***	0.120***	0.097***
	(9.38)	(6.29)	(9.46)	(6.53)
Year2022	0.085***	0.057***	0.085***	0.059***
	(7.12)	(3.35)	(6.91)	(3.52)
ln(GDP per capita)		0.085*		0.051
		(1.84)		(1.10)
GDP growth		−0.054		−0.093
		(−0.30)		(−0.47)
Fiscal deficit		−0.322		−0.550**
		(−1.29)		(−2.04)
Tax ratio		−0.025		−0.024
		(−0.31)		(−0.28)
Secondary sector		−0.948**		−1.097**
		(−2.13)		(−2.44)
Tertiary sector		−0.555		−0.747*
		(−1.23)		(−1.66)
Constant	0.110***	−0.113	0.105***	0.441
	(18.89)	(−0.21)	(18.53)	(0.88)
Observations	1,374	1,324	1,374	1,324
Adj. R-squared	0.451	0.453	0.412	0.415
City FE	YES	YES	YES	YES

Table 5
DID analysis of land price and local government fiscal conditions

	(1)	(2)	(3)	(4)	(5)
VARIABLES			*ln*(*land price*)		
*Land dependence_2019 * year2015*	0.331	0.112			0.090
	(1.52)	(0.58)			(0.47)
*Land dependence_2019 * year2016*	0.267	0.086			0.091
	(1.26)	(0.40)			(0.48)
*Land dependence_2019 * year2017*	0.184	0.165			0.131
	(0.93)	(0.91)			(0.74)
*Land dependence_2019 * year2018*	0.113	0.058			0.046
	(0.70)	(0.40)			(0.32)
*Land dependence_2019 * year2020*	0.234	0.087			0.042
	(1.51)	(0.62)			(0.29)
*Land dependence_2019 * year2021*	0.289*	0.175			0.080
	(1.79)	(1.20)			(0.55)
*Land dependence_2019 * year2022*	0.433**	0.369**			0.271*
	(2.51)	(2.32)			(1.66)
*LGFV debt_2019 * year2015*			0.025	0.010	0.006
			(1.21)	(0.53)	(0.27)
*LGFV debt_2019 * year2016*			–0.001	0.004	–0.002
			(–0.03)	(0.16)	(–0.09)
*LGFV debt_2019 * year2017*			0.010	0.017	0.014
			(0.49)	(1.06)	(0.76)
*LGFV debt_2019 * year2018*			0.000	0.000	0.001
			(0.00)	(0.00)	(0.10)
*LGFV debt_2019 * year2020*			0.015	0.015	0.017
			(1.15)	(1.31)	(1.30)
*LGFV debt_2019 * year2021*			0.030	0.036***	0.034**
			(1.53)	(2.58)	(2.12)
*LGFV debt_2019 * year2022*			0.037**	0.043***	0.037**
			(2.13)	(2.80)	(2.06)
Constant	9.453***	7.363***	9.141***	7.026***	7.489***
	(5.86)	(5.64)	(7.14)	(6.27)	(5.60)
Observations	121,216	109,354	129,926	117,442	109,354
Adj. R-squared	0.384	0.539	0.388	0.539	0.539
City controls	YES	YES	YES	YES	YES
Transaction controls	NO	YES	NO	YES	YES
Year/quarter FE	YES	YES	YES	YES	YES
City FE	YES	YES	YES	YES	YES

Interpretation (Table 4).

- Failure ratios increase sharply in 2020–2022, especially 2021–2022, indicating more failed auctions (both by count and by area).
- This matters because it suggests that the decline in land-market activity is not *only* due to local governments reducing supply; demand weakness shows up via failed auctions.

Interpretation (Table 5).

- **Land dependence**: the interaction becomes significantly positive in 2022 (and marginally in 2021), meaning land prices rise more in cities that relied more on land revenue pre-COVID.
- **LGFV debt**: the interaction is significantly positive in 2021–2022 (especially with transaction controls), consistent with stronger price support in more indebted cities.
- Overall: the cross-city heterogeneity lines up with the incentive story: fiscal stress and land-linked debt capacity predict stronger price support.

Table 6
Divergences in land prices and transaction volumes across cities

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	Land price growth rate						
Variables	All	*Land dependence_2019*			*LGFV debt_2019*		
		Low	Middle	High	Low	Middle	High
Land area growth rate	0.034	–0.007	–0.029	0.143	0.018	0.103	–0.013
	(0.81)	(–0.13)	(–0.44)	(1.41)	(0.24)	(1.39)	(–0.21)
*Land area growth rate * Covid19*	–0.145**	–0.126	–0.055	–0.166	–0.092	–0.201*	–0.160**
	(–2.50)	(–1.45)	(–0.63)	(–1.36)	(–0.83)	(–1.98)	(–2.05)
Constant	0.678	3.300*	3.629	–3.825	2.450	1.566	–1.924
	(0.61)	(1.86)	(1.19)	(–1.58)	(1.18)	(1.01)	(–0.76)
Observations	1,325	394	392	389	445	447	433
Adj. R-squared	0.068	0.059	0.058	0.030	0.057	0.043	0.106
City controls	YES	YES	YES	YES	YES	YES	YES
City FE	YES	YES	YES	YES	YES	YES	YES
Year FE	YES	YES	YES	YES	YES	YES	YES

My interpretation (Table 6).

- Pre-COVID, price growth and volume growth are basically uncorrelated (a_1 near 0).
- During COVID, the correlation turns negative ($a_2 < 0$), i.e., cities with larger declines in land volume experience *higher* land-price growth (divergence).
- The negative comovement is strongest (and statistically significant) in the **high LGFV-debt** group (col. 7), consistent with more active price support in more indebted cities.

Interpretation (Table 7).

- LGFVs dramatically increase their land purchases in 2021 and (especially) 2022: the coefficients jump to 0.060 and 0.195 in panel A (area ratio) and 0.055 and 0.211 in panel B (value ratio).
- The increases are **stronger in high land-dependence cities** and **high LGFV-debt cities**, matching the incentive mechanism: these places have more to gain from supporting land prices.

Interpretation (Table 8).

- The baseline LGFV dummy is negative (about -0.08 to -0.09): in the omitted year (2019), LGFVs pay lower prices than other buyers, consistent with preferential access or targeting cheaper parcels.
- During COVID, LGFVs bid **more aggressively**: the interaction terms are positive in 2020 and especially in 2022 (0.117 in col. 3), implying higher relative prices paid by LGFVs in those years.
- LGFVs both *buy more* land and (in some years) *pay relatively higher prices*—a strong footprint of policy-driven support.

Interpretation (Table 9).

- LGFV leverage rises strongly in 2021–2022 (about +3.7 and +4.1 percentage points relative to 2019).
- This is consistent with LGFVs taking on more balance-sheet risk to support land purchases (and possibly local stimulus projects) during the downturn.

Interpretation (Table 10).

- **Housing prices rise in 2020–2022** (positive year dummies), even as quantities fall.
- Volumes collapse in 2022: transaction units/area/value all show large negative coefficients (around -0.38 to -0.41 log points).
- This replicates the land-market pattern: price support alongside reduced market turnover.

Interpretation (Table 11).

- In high-LGFV-debt cities, housing prices rise more in 2021–2022 (positive and significant interactions in col. 1).
- In 2022, those same cities issue significantly fewer sales permits (negative and significant

Table 7
Land purchases by LGFVs
A. LGFV land ratio (area)

Variables	(1)	(2)	(3)	(4)	(5)	(6)	(7)
				LGFV land ratio (area)			
	All	*Land dependence_2019*			*LGFV debt_2019*		
		Low	Middle	High	Low	Middle	High
Year2015	0.002	0.020	−0.013	−0.116*	−0.025	0.062**	−0.066
	(0.10)	(0.78)	(−0.48)	(−2.00)	(−0.91)	(2.04)	(−1.45)
Year2016	0.003	0.031	−0.011	−0.091**	−0.019	0.019	−0.021
	(0.17)	(1.20)	(−0.44)	(−2.09)	(−0.78)	(0.72)	(−0.56)
Year2017	−0.004	0.039*	−0.027	−0.072**	−0.007	0.039	−0.056**
	(−0.27)	(1.97)	(−1.41)	(−2.25)	(−0.35)	(1.62)	(−2.06)
Year2018	−0.002	−0.010	−0.002	−0.014	0.003	−0.001	−0.009
	(−0.16)	(−0.73)	(−0.17)	(−0.58)	(0.23)	(−0.06)	(−0.40)
Year2020	0.033**	0.028	0.039	0.046*	0.037*	0.003	0.054*
	(2.36)	(1.19)	(1.44)	(1.89)	(1.80)	(0.16)	(1.98)
Year2021	0.060***	0.065***	0.023	0.127***	0.051***	0.049**	0.088***
	(4.99)	(3.19)	(1.06)	(4.82)	(3.02)	(2.29)	(3.59)
Year2022	0.195***	0.180***	0.182***	0.269***	0.175***	0.156***	0.258***
	(10.66)	(6.43)	(5.54)	(6.43)	(5.43)	(4.62)	(9.25)
Constant	−0.565	−2.075**	−0.984	3.823**	−0.188	−1.548	0.896
	(−0.80)	(−2.23)	(−0.64)	(2.56)	(−0.22)	(−1.22)	(0.60)
Observations	1,332	401	392	389	446	453	433
Adj. R-squared	0.572	0.438	0.627	0.607	0.438	0.539	0.550
City controls	YES	YES	YES	YES	YES	YES	YES
City FE	YES	YES	YES	YES	YES	YES	YES

B. LGFV land ratio (valuation)

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
				LGFV land ratio (value)			
Variables	All	*Land dependence_2019*			*LGFV debt_2019*		
Low	Middle	High	Low	Middle	High		
Year2015	0.034*	0.063***	−0.016	−0.063	0.019	0.088***	−0.040
	(1.69)	(2.71)	(−0.55)	(−1.16)	(0.75)	(2.86)	(−0.89)
Year2016	0.018	0.052*	−0.013	−0.052	0.005	0.024	−0.005
	(1.05)	(1.97)	(−0.48)	(−1.23)	(0.21)	(0.89)	(−0.12)
Year2017	−0.004	0.051**	−0.041**	−0.063**	0.003	0.028	−0.058**
	(−0.34)	(2.49)	(−2.13)	(−2.01)	(0.16)	(1.16)	(−2.06)
Year2018	−0.003	0.002	−0.010	−0.021	0.002	−0.004	−0.008
	(−0.28)	(0.12)	(−0.70)	(−0.93)	(0.18)	(−0.22)	(−0.38)
Year2020	0.032**	0.024	0.043	0.046	0.037*	0.003	0.047
	(2.23)	(1.02)	(1.58)	(1.57)	(1.69)	(0.16)	(1.67)
Year2021	0.055***	0.057**	0.037	0.110***	0.041**	0.056**	0.077***
	(4.38)	(2.66)	(1.61)	(4.03)	(2.45)	(2.40)	(2.96)
Year2022	0.211***	0.190***	0.215***	0.288***	0.194***	0.167***	0.277***
	(10.56)	(6.01)	(5.67)	(6.46)	(5.51)	(4.48)	(9.15)
Constant	−0.643	−2.080**	−1.209	3.315**	−0.700	−1.294	0.786
	(−0.95)	(−2.26)	(−0.78)	(2.45)	(−0.97)	(−1.05)	(0.52)
Observations	1,332	401	392	389	446	453	433
Adj. R-squared	0.550	0.415	0.569	0.609	0.414	0.571	0.511
City controls	YES	YES	YES	YES	YES	YES	YES
City FE	YES	YES	YES	YES	YES	YES	YES

Table 8
Price of land purchased by LGFVs

VARIABLES	(1) *ln(land price)*	(2) *ln(land price)*	(3) *ln(land price)*
LGFV dummy	−0.082**	−0.082**	−0.090**
	(−2.06)	(−2.05)	(−2.52)
*LGFV dummy * year2015*	0.239***	0.240***	0.173***
	(4.64)	(4.63)	(3.84)
*LGFV dummy * year2016*	0.030	0.028	−0.016
	(0.43)	(0.40)	(−0.21)
*LGFV dummy * year2017*	−0.089	−0.085	−0.077
	(−1.59)	(−1.52)	(−1.35)
*LGFV dummy * year2018*	−0.003	−0.004	−0.043
	(−0.05)	(−0.08)	(−0.87)
*LGFV dummy * year2020*	0.075*	0.081**	0.070*
	(1.94)	(2.08)	(1.95)
*LGFV dummy * year2021*	0.055	0.057	0.064
	(1.14)	(1.16)	(1.59)
*LGFV dummy * year2022*	0.157***	0.154***	0.117***
	(3.34)	(3.27)	(2.82)
ln(land area)			0.082***
			(6.47)
FAR			0.241***
			(24.59)
Urban			0.522***
			(19.83)
Land grade			−0.121***
			(−9.51)
New land			−0.021***
			(−8.39)
Tender			0.039
			(0.39)
Auction			0.257***
			(9.37)
Constant	7.899***	9.005***	6.818***
	(464.76)	(6.94)	(6.31)
Observations	133,965	129,926	117,442
Adj. R-squared	0.387	0.389	0.539
City controls	NO	YES	YES
Year/quarter FE	YES	YES	YES
City FE	YES	YES	YES

Table 9
LGFV leverage

VARIABLES	*LGFV leverage* (%)
Year2015	–2.951**
	(–2.03)
Year2016	0.106
	(0.10)
Year2017	1.481**
	(2.05)
Year2018	–0.111
	(–0.27)
Year2020	0.858*
	(1.74)
Year2021	3.693***
	(4.83)
Year2022	4.061***
	(4.05)
Constant	–10.846
	(–0.28)
Observations	1,279
Adj. R-squared	0.750
City controls	YES
City FE	YES

Table 10
Price and volume of transactions of new housing units

VARIABLES	(1) *ln(house price)*	(2) *ln(house transaction unit)*	(3) *ln(house transaction area)*	(4) *ln(house transaction value)*
Year2015	–0.343***	0.246	0.223	0.089
	(–9.76)	(1.54)	(1.37)	(0.51)
Year2016	–0.347***	0.131	0.094	–0.169
	(–14.21)	(1.37)	(0.98)	(–1.63)
Year2017	–0.210***	–0.123	–0.143*	–0.288***
	(–10.38)	(–1.42)	(–1.66)	(–3.25)
Year2018	–0.072***	–0.075	–0.084*	–0.123**
	(–6.11)	(–1.48)	(–1.66)	(–2.36)
Year2020	0.030**	0.073	0.071	0.075
	(2.42)	(1.26)	(1.23)	(1.21)
Year2021	0.059***	–0.119*	–0.115*	–0.051
	(4.02)	(–1.78)	(–1.73)	(–0.74)
Year2022	0.054**	–0.414***	–0.394***	–0.379***
	(2.34)	(–4.09)	(–3.91)	(–3.62)
Constant	9.103***	4.973	9.922**	6.488
	(10.91)	(1.23)	(2.44)	(1.53)
Observations	1,013	1,014	1,014	1,033
Adj. R-squared	0.943	0.761	0.757	0.823
City controls	YES	YES	YES	YES
City FE	YES	YES	YES	YES

Table 11
DID analysis of new house price and sales permits

VARIABLES	(1) *ln(house price)*	(2) *ln(permit area)*	(3) *ln(permit units)*
*LGFV debt_2019 * year2015*	–0.001	–0.015	0.001
	(–0.08)	(–0.14)	(0.01)
*LGFV debt_2019 * year2016*	0.002	–0.153*	–0.114
	(0.17)	(–1.72)	(–1.12)
*LGFV debt_2019 * year2017*	–0.011	–0.092	–0.104
	(–1.17)	(–0.93)	(–0.92)
*LGFV debt_2019 * year2018*	–0.011	–0.048	–0.055
	(–1.51)	(–0.71)	(–0.80)
*LGFV debt_2019 * year2020*	0.006	–0.023	–0.020
	(1.21)	(–0.42)	(–0.37)
*LGFV debt_2019 * year2021*	0.018**	0.017	0.012
	(2.08)	(0.31)	(0.21)
*LGFV debt_2019 * year2022*	0.022**	–0.132**	–0.143**
	(2.01)	(–2.06)	(–1.99)
ln(house price)		0.847**	0.809**
		(2.52)	(2.25)
ln(house transaction area t-1)		0.118	
		(1.19)	
ln(house transaction units t-1)			0.084
			(0.66)
Constant	9.390***	–5.952	–16.970
	(11.25)	(–0.41)	(–1.04)
Observations	1,005	564	563
Adj. R-squared	0.942	0.704	0.693
City controls	YES	YES	YES
Year FE	YES	YES	YES
City FE	YES	YES	YES

Table 12
Land price, firm financing cost and city investment

Variables	(1) *Coupon spread All firms*	(2) *Coupon spread All firms*	(3) *Coupon spread Non-LGFV*	(4) *Coupon spread LGFV*
ln(land price avgq)	−0.053*	−0.016	0.024	−0.071
	(−1.70)	(−0.45)	(0.86)	(−0.82)
*ln(land price avgq) * covid19*		−0.063**	−0.108***	0.025
		(−2.25)	(−3.34)	(0.34)
ln(issue amount)	−0.041**	−0.040**	−0.043**	−0.028
	(−2.45)	(−2.38)	(−2.46)	(−1.63)
Maturity	−0.030**	−0.030**	−0.014	−0.039***
	(−2.44)	(−2.43)	(−0.85)	(−2.74)
TripleA	−0.481***	−0.494***	−0.288	−0.579***
	(−4.45)	(−4.52)	(−1.49)	(−5.24)
DoubleAplus	−0.181**	−0.193**	0.046	−0.316***
	(−2.00)	(−2.10)	(0.24)	(−5.03)
SOE	−0.479***	−0.479***	−0.480***	
	(−5.08)	(−4.75)	(−3.66)	
ln(asset)	−0.086	−0.071	0.087	−0.317
	(−0.58)	(−0.47)	(0.71)	(−1.42)
ln(sales)	−0.037	−0.043	−0.105**	0.009
	(−1.00)	(−1.20)	(−2.11)	(0.20)
Leverage	0.616	0.573	0.658	0.420
	(1.35)	(1.26)	(1.39)	(1.05)
ROA	−2.086*	−2.222*	−2.038*	1.831
	(−1.69)	(−1.79)	(−1.78)	(0.60)
Constant	19.134***	18.573***	5.025	38.464***
	(2.69)	(2.64)	(0.80)	(3.21)
Observations	29,246	29,246	19,002	10,225
Adj. R-squared	0.808	0.808	0.821	0.782
City controls	YES	YES	YES	YES
Year/quarter FE	YES	YES	YES	YES
Firm FE	YES	YES	YES	YES
Bond type FE	YES	YES	YES	YES

Table 13
DID analysis of developers' housing sales

VARIABLES	(1) *ln(house sales value)*	(2) *ln(House sales area)*
*LGFV debt_2019 * year2015*	−0.034	−0.034
	(−0.91)	(−0.96)
*LGFV debt_2019 * year2016*	0.008	0.022
	(0.22)	(0.63)
*LGFV debt_2019 * year2017*	−0.017	−0.006
	(−0.45)	(−0.16)
*LGFV debt_2019 * year2018*	0.029	0.041
	(1.01)	(1.47)
*LGFV debt_2019 * year2020*	−0.008	−0.006
	(−0.35)	(−0.25)
*LGFV debt_2019 * year2021*	−0.008	−0.009
	(−0.27)	(−0.33)
*LGFV debt_2019 * year2022*	−0.074**	−0.073**
	(−2.17)	(−2.11)
Constant	11.366***	10.731***
	(2.95)	(2.80)
Observations	16,171	16,171
Adj. R-squared	.306	.262
Developer controls	YES	YES
City controls	YES	YES
Year FE	YES	YES
Developer FE	YES	YES
City FE	YES	YES

interactions in cols. 2–3).

- This strongly supports the **supply restriction channel**: local governments curb new supply (permits) to prop up prices.

Interpretation (Table 12).

- During COVID, higher land prices predict **lower financing costs** for **non-LGFV firms** (negative interaction in col. 3).
- For LGFV bonds (col. 4), the interaction is small and insignificant, consistent with pricing being dominated by implicit government backing rather than local collateral values.
- This table ties land-price support to broader local credit conditions (a key "why would local governments do this?" piece).

Interpretation (Table 13).

- In 2022, developers operating in high-LGFV-debt cities sell significantly less (both value and area), consistent with tighter local controls (e.g., permit restrictions) and weaker market liquidity.
- This is an important *real-side consequence*: price support via supply restrictions can squeeze developers' cash flows.

5 Conclusion and summary

Bottom line / conclusion. During the COVID-19 downturn (roughly 2020–2022), Chinese cities experienced an unusual and sustained *price–volume divergence* in both the **primary land market** and the **new-housing market**: prices stayed high (even rose) while transaction volumes fell sharply. The paper concludes that this pattern is *not* well explained by standard real-estate frictions (e.g., households' loss aversion, search frictions, speculative dynamics) because the key sellers in China are *institutions*—local governments in the land market and developers in the new-home market. Instead, the evidence points to **active local-government price management**, motivated by local governments' fiscal reliance on land and their debt financing that is tied to land/real-estate collateral.

Summary (what the paper does and what it finds).

- **Fact to explain:** In the cross-section of Chinese cities, land (and new-home) prices rose or stayed elevated in 2021–2022, even as transaction volumes contracted substantially. The divergence is visible in aggregate time series and remains after controlling for city and parcel characteristics in panel regressions.
- **Core hypothesis:** Local governments have strong incentives to prevent land/housing prices from falling in downturns because: (i) land sales are a major revenue source, and (ii) land and real estate values directly support debt financing capacity (especially via LGFVs), since collateral values are based on observed market prices.
- **Empirical strategy:** The paper treats COVID-19 as a large negative shock and uses dynamic DID-style specifications to compare post-2020 price dynamics across cities with different *pre-pandemic* (2019) exposure to land-based finance:
  1. *Land dependence* (share of land-sale revenue in total fiscal revenue in 2019), and
  2. *LGFV debt burden* (city-level LGFV liabilities scaled by total fiscal revenue in 2019).

  Pre-trends are broadly flat (supporting parallel trends), while post-2020 effects load strongly on these 2019 measures.
- **Main results:**
  1. Cities with higher 2019 land dependence and, especially, higher 2019 LGFV debt experienced **larger land-price increases** in 2021–2022 (relative to 2019).
  2. Across cities, the price–volume relationship becomes **more negative** during COVID, with the divergence strongest in **high-LGFV-debt** cities.
- **Mechanisms (how price management shows up in data):**
  1. **LGFVs as marginal buyers:** LGFVs sharply increased their share of land purchases during the pandemic (countercyclical buying). Moreover, LGFVs switched from paying *lower* prices than other bidders pre-pandemic to paying *higher* relative prices by 2022, consistent with "buyer of last resort" behavior that props up auction prices.
  2. **Housing-market supply control:** In the new-home market, cities with heavier LGFV-debt burdens issued **fewer sales permits** in 2022 (restricting supply), while also exhibiting larger housing-price increases in 2021–2022. In addition, the paper documents administrative rules in some cities that restrict developers from cutting prices below benchmarks.
- **Consequences: a dual-edged effect.**
  1. *Potentially stabilizing channel:* Higher land prices are associated with **lower financing costs** (bond spreads) for local nonfinancial firms during the pandemic, consistent with

a collateral / signal channel in credit markets.

2. *Destabilizing channel:* Developers' housing sales fall **more** in cities with heavier pre-pandemic LGFV debt, consistent with the idea that keeping prices "too high" (or constraining supply via permits/price floors) can choke off transactions and intensify developers' liquidity stress.

How we would explain the mechanism in one story. Think of local governments as a **monopolistic land seller** whose balance sheet (via LGFVs) is effectively **long real estate collateral values**. When debt is large and rollover risk matters, a local government may prefer to defend the *price level* even if it sacrifices *quantity*: high observed land/housing prices support collateral values and borrowing capacity, while a sharp price drop can trigger a negative spiral (worse collateral → tighter credit → weaker local activity → even worse prices). In that environment, it can be rational (from the local government's perspective) to accept "fewer transactions at a higher price".
A simple way I summarize the logic is:

$$\text{Local fiscal capacity} \approx \underbrace{P \cdot Q}_{\text{land-sale cash flow}} + \underbrace{\Phi(P)}_{\text{borrowing capacity via collateral}},$$

where $\Phi(P)$ is increasing in the observable market price P (because collateral and market-value-based risk rules scale with prices). When the marginal value of collateral is high (high debt, refinancing pressure), the government places more weight on sustaining P than on sustaining Q. Operationally, this shows up as reserve-price discipline, supply control, and (crucially) LGFVs stepping in to bid/buy, which mechanically supports auction outcomes and price indices, even as private demand weakens.

Takeaway The paper reframes China's real-estate downturn as not only a developer/speculation story, but also a **local-public-finance** story: credit supply and fiscal financing are intertwined with land and housing prices, so local governments have incentives to intervene directly in market prices. This creates a feedback loop: financial institutions lend against real-estate values ⇒ local governments want to prop up those values ⇒ market signals become distorted. The conclusion is that **developer bailouts alone cannot fix the root problem**; a durable solution requires **reforming local public finance** to reduce reliance on land sales and land-collateralized (LGFV) debt as core financing tools.